

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION 1

YEAR 11 COMMON TEST

MAY 2005

Time allowed: 70 minutes

Instructions:

- Show all necessary working in every question
- Start each question on a new page
- Attempt all questions
- All questions are not of equal value
- Marks shown are approximate & may be changed
- Full marks may not be awarded for careless or badly arranged work
- Your sketches must be neat. Use a ruler to draw axes.
- Approved calculators may be used
- These questions are to be handed in with your answers.

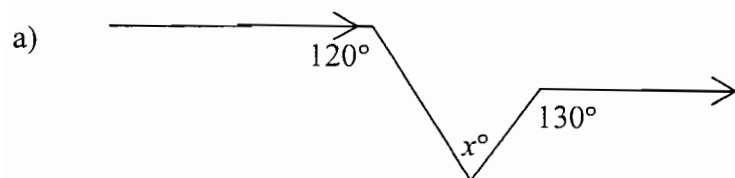
Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/10	/8	/8	/8	/54

Question 1

Marks



Find the value of x (no reasons necessary).

1

b) Factorise $a^2 - b^2 - (a - b)^2$

2

c) The hyperbola $y = \frac{3}{a - x}$ has a vertical asymptote at $x = 1$. What is the value of a ?

1

d) If $\tan a = -\frac{1}{3}$ and $\cos a > 0$, find the exact value of $\sin a$.

2

e) Given that n is a positive number indicate

2

(i) the largest

(ii) the smallest

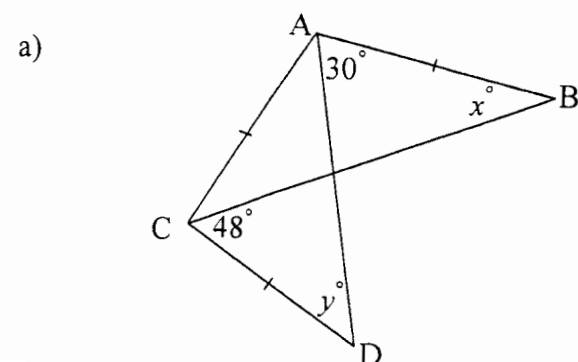
of the following numbers:

$$3^{-\frac{n}{3}}, 3^{\frac{n}{3}}, 3^n, 3^{-n}$$

f) Solve $|5x - 3| = |3x + 1|$

2

Question 2 (start a new page)



$\triangle ABC$ and $\triangle ACD$ are isosceles.

3

By forming a pair of simultaneous equations or otherwise, find the value of x .

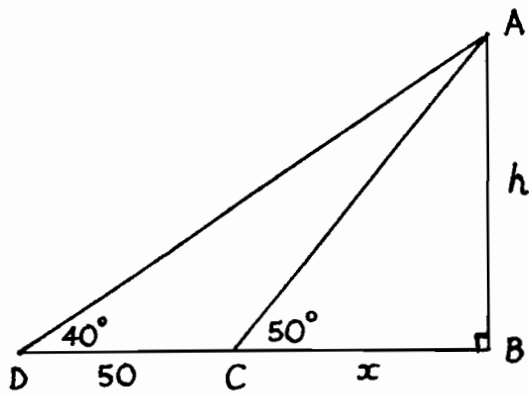
- b) If $\frac{a^n + a^{n+2}}{a^n} = 10$ find a given that $a > 0$. 2
- c) Solve $\frac{x-2}{x} \geq 1$ 3
- d) There are two values of θ in the domain $0^\circ \leq \theta \leq 360^\circ$ where $\sin \theta$ and $\cos \theta$ are numerically equal. Find these two values. 2

Question 3 (start a new page)

- a) If $p = \frac{\sqrt{3}}{4 - \sqrt{3}}$ and $q = \frac{\sqrt{3}}{4 + \sqrt{3}}$ evaluate $\frac{p+q}{1-pq}$ 3
- b) How many solutions does the equation $(\cos x - 2)(\sin^2 x - 1) = 0$ have in the domain $0^\circ \leq x \leq 360^\circ$? 3
There is no need to solve the equation.
Justify your answer.
- c) i) Sketch the graph of $y = x^2 + \frac{1}{2}$ 4
- ii) On a separate diagram sketch $y = \frac{2}{2x^2 + 1}$.
- iii) Use your diagram or otherwise to write down the range of $y = \frac{2}{2x^2 + 1}$

Question 4 (start a new page)

a)



We wish to find the height AB of a vertical cliff. From a point D the angle of elevation of A is 40° . From a point C 50m nearer the base of the cliff the angle of elevation is 50° .

4

- i) Show that $h = (50 + x) \tan 40^\circ$
- ii) Show that $h = x \tan 50^\circ$
- iii) Using simultaneous equations find h

b) i) Sketch the graph of $y = |x + 1|$.

4

- ii) By using your graph or otherwise solve $\frac{2}{x} > |x + 1|$

Question 5 (start a new page).a) i) Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$

4

- ii) Hence solve $-\frac{\sqrt{3}}{2} \leq \cos x \leq \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$

b) i) If $xy = c^2$ prove that $\frac{1}{c+x} + \frac{1}{c+y} = \frac{1}{c}$.

4

- ii) Hence or otherwise simplify $\frac{1}{6 + \sqrt{51} + \sqrt{15}} + \frac{1}{6 + \sqrt{51} - \sqrt{15}}$

Question 6 (start a new page)

- a) Find the point/s of intersection for the graphs of $y = x^2 - 1$ and $y = \frac{1}{x^2 - 1}$ 3
- b) i) Sketch the graph of $y = -\sqrt{2 - x^2}$. 5
- ii) On the same diagram shade the region where $y \geq -\sqrt{2 - x^2}$, $|x| \leq 1$ and $y \leq 0$ hold simultaneously.
- iii) Find the exact value for the area of the shaded region.

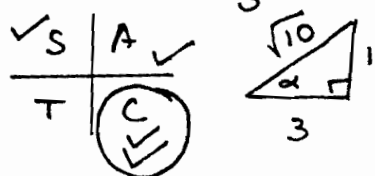
Question 1

a) $x = 70$

b) $a^2 - b^2 - (a-b)^2$
 $(a-b)(a+b) - (a-b)^2$
 $(a-b)((a+b) - (a-b))$
 $2b(a-b)$

c) $y = \frac{3}{a-x}$ $a-x=0$
 $\therefore \underline{a=1}$

d) $\tan \alpha = -\frac{1}{3}$ $\cos \alpha > 0$



$\therefore \underline{\sin \alpha = -\frac{1}{\sqrt{10}}}$

e) $3^{-\frac{n}{3}}, 3^{\frac{n}{3}}, 3^n, 3^{-n}$
 i) largest 3^n
 ii) smallest 3^{-n} } n a +ve integer

f) $|5x-3| = |3x+1|$

$5x-3 = 3x+1$ or $5x-3 = -3x-1$

$2x = 4$

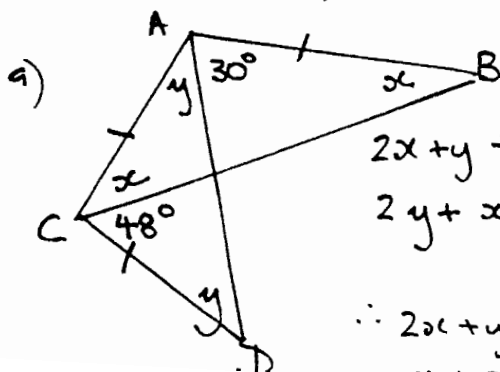
$8x = 2$

$\underline{x=2}$

OR

$\underline{x = \frac{1}{4}}$

Question 2



$2x + y + 30 = 180$

$2y + x + 48 = 180$

$\therefore 2x + y = 150 \text{ --- ①}$

$x + 2y = 132 \text{ --- ②}$

b) $\frac{a^n + a^{n+2}}{a^n} = 10$

$\cancel{a^n} (1 + a^2) = 10$

$\therefore a^2 = 9$

$\underline{a = 3 \text{ only } a > 0}$

c) $\frac{x-2}{x} \geq 1$

$x(x-2) \geq x^2$

$x(x-2) - x^2 \geq 0$

$x(\cancel{x} - 2 - \cancel{x}) \geq 0$

$-2x \geq 0$

$x \leq 0$ but $x \neq 0$

$\therefore \underline{x < 0}$

d) $\sin \theta = \cos \theta$

$\tan \theta = 1$

acute $\theta = 45$

$\therefore \underline{\theta = 45^\circ, 225^\circ}$

Question 3

a) $p = \frac{\sqrt{3}}{4-\sqrt{3}} \times \frac{4+\sqrt{3}}{4+\sqrt{3}} = \frac{4\sqrt{3}+3}{13}$

$q = \frac{\sqrt{3}}{4+\sqrt{3}} \times \frac{4-\sqrt{3}}{4-\sqrt{3}} = \frac{4\sqrt{3}-3}{13}$

$p+q = \frac{4\sqrt{3}+3+4\sqrt{3}-3}{13}$
 $= \frac{8\sqrt{3}}{13}$

$1-pq = 1 - \frac{\sqrt{3}}{4-\sqrt{3}} \times \frac{\sqrt{3}}{4+\sqrt{3}}$

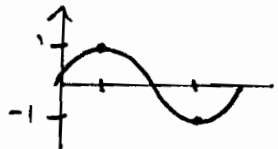
$$\frac{p+q}{1-pq} = \frac{8\sqrt{3}}{13} \div \frac{10}{13}$$

$$= \frac{8\sqrt{3}}{10}$$

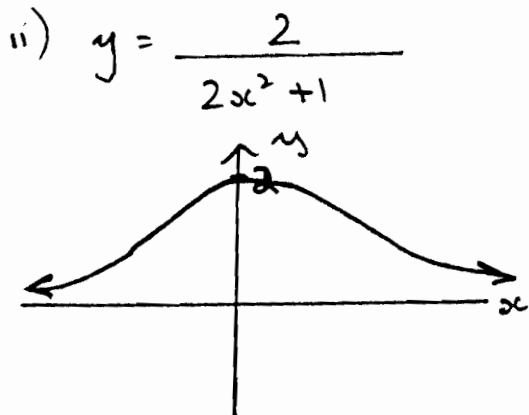
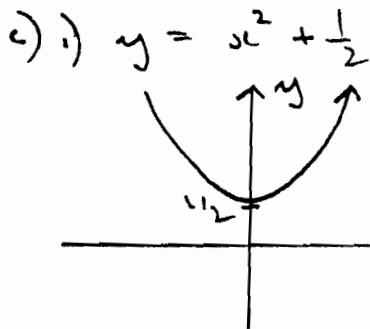
$$= \frac{4\sqrt{3}}{5}$$

b) $(\cos x - 2)(\sin^2 x - 1) = 0$

$\cos x = 2$ $\sin^2 x = 1$
no solutions $\sin x = \pm 1$



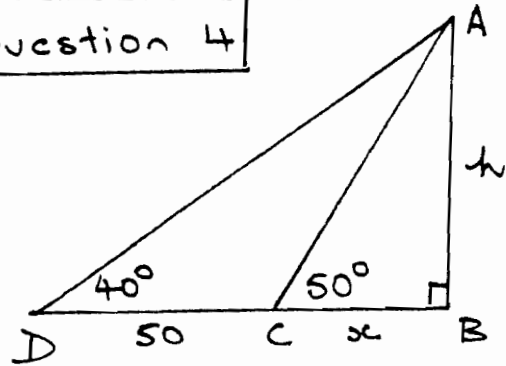
2 solutions



iii) Range $0 < y \leq 2$

Question 4

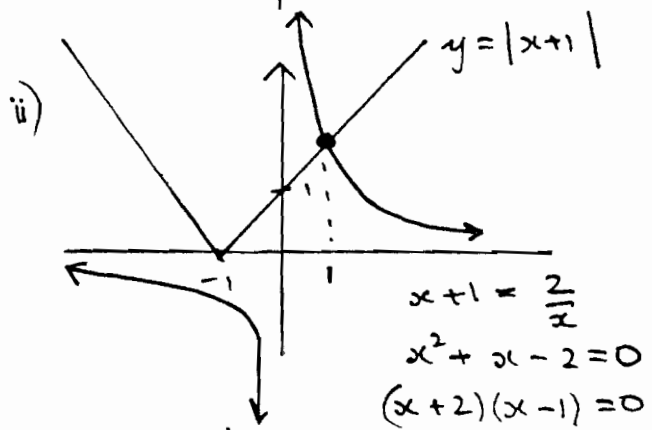
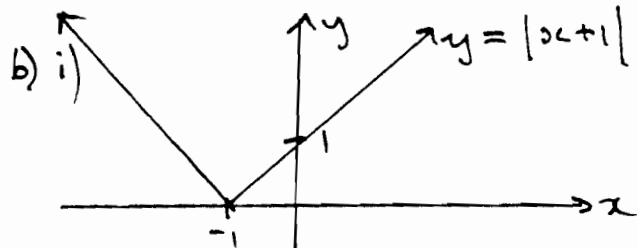
a)



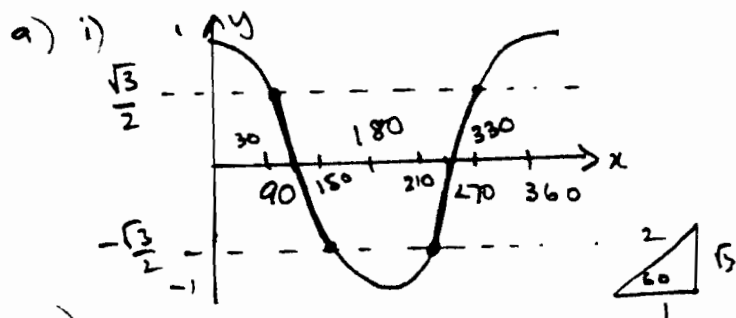
ii) $\tan 50^\circ = \frac{h}{x}$
 $\therefore h = x \tan 50^\circ$

i) $\tan 40^\circ = \frac{h}{50+x}$
 $(50+x) \tan 40^\circ = h$

iii) $x \tan 50^\circ = (50+x) \tan 40^\circ$
 $x \tan 50^\circ = 50 \tan 40^\circ + x \tan 40^\circ$
 $x \tan 50^\circ - x \tan 40^\circ = 50 \tan 40^\circ$
 $x (\tan 50^\circ - \tan 40^\circ) = 50 \tan 40^\circ$
 $x = \frac{50 \tan 40^\circ}{\tan 50^\circ - \tan 40^\circ}$
 $x = 118.97 \text{ units (2 dec pl)}$



Question 5



ii)

$$\cos x = \frac{\sqrt{3}}{2}, \quad \cos x = -\frac{\sqrt{3}}{2}$$

acute $x > 30^\circ$

$$\underline{\underline{x = 30^\circ, 150^\circ}} \quad \underline{\underline{x = 210^\circ, 330^\circ}}$$

$\therefore$ Solution: $30^\circ \leq x \leq 150^\circ$
and $210^\circ \leq x \leq 330^\circ$

b)

$$\frac{1}{c+x} + \frac{1}{c+y}$$

$$= \frac{c+y + c+x}{(c+x)(c+y)}$$

$$= \frac{2c + x + y}{c^2 + cx + cy + xy}$$

$$= \frac{2c + x + y}{2c^2 + cx + cy}$$

$$= \frac{(2c + x + y)}{c(2c + x + y)}$$

$$= \frac{1}{c}$$

$$\therefore \underline{\underline{\frac{1}{c+x} + \frac{1}{c+y} = \frac{1}{c}}}$$

ii) If $x = \sqrt{51} + \sqrt{15}$

$$y = \sqrt{51} - \sqrt{15}$$

$$\frac{1}{6+x} + \frac{1}{6+y} = \underline{\underline{\frac{1}{6}}}$$

Question 6

a) $y = x^2 - 1$ $y = \frac{1}{x^2 - 1}$

$$x^2 - 1 = \frac{1}{x^2 - 1}$$

$$(x^2 - 1)^2 = 1$$

$$x^2 - 1 = 1 \quad \text{or} \quad x^2 - 1 = -1$$

$$x^2 = 2$$

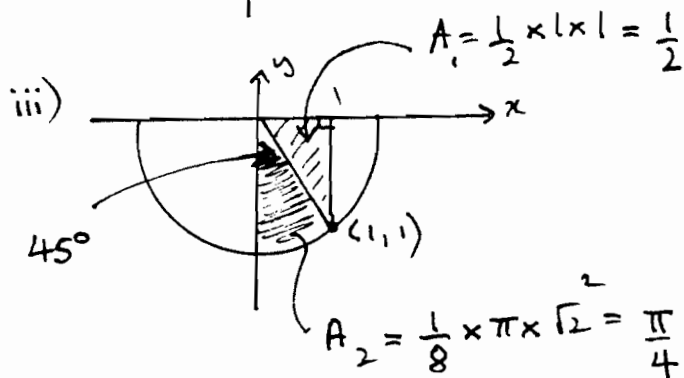
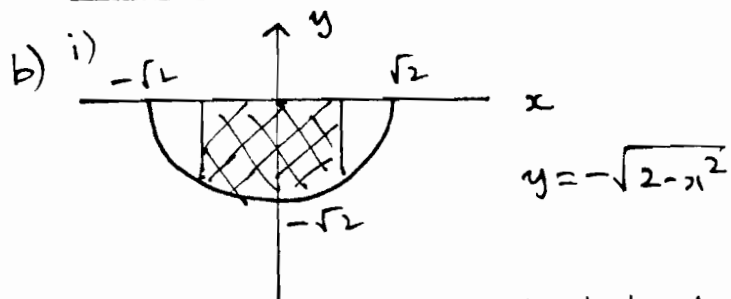
$$x^2 = 0$$

$$x = \pm\sqrt{2}$$

$$x = 0$$

$\therefore$ pts intersection

$$\underline{\underline{(\sqrt{2}, 1) \quad (-\sqrt{2}, 1) \quad (0, -1)}}$$



$\therefore$ Required area = $2 \left(\frac{1}{2} + \frac{\pi}{4} \right)$